KAYCEE NDUKUBA

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PROFESSIONAL SUMMARY

Data Analyst (MSc Data Science) who turns messy, real-world data into clear business insight with SQL, Python and rigorous analysis. Built a portfolio of applied analytics and machine learning projects on real-world datasets spanning customer analytics, behavioural modelling, NLP and retrieval-augmented AI, with a focus on translating technical outputs into business decisions. Strong in reporting, ETL and stakeholder communication, and a confident communicator with a psychology background — backed by customer-analytics projects with quantified impact (churn, lifetime value, recommendations).

CORE SKILLS

Querying & Data: SQL (PostgreSQL, BigQuery, Snowflake), Python (pandas, NumPy), Excel, R
Analytics & Reporting: Exploratory data analysis, KPI/metric design, segmentation, cohort & customer analytics, dashboards, stakeholder reporting
Data Engineering: ETL pipelines, dbt, automation, data quality & validation, Git/GitHub
Statistics & ML: Hypothesis testing, regression & classification, feature engineering, model evaluation, SHAP interpretability
Visualisation: Matplotlib, Seaborn, Jupyter (transferable to Tableau / Power BI)

EXPERIENCE

AI & Data Scientist — Yves OS, London *Dec 2025 – Present*

- Developed behavioural analytics workflows that transformed multi-platform user-interaction data into measurable insights, used to prioritise product features and improve user experience.
- Applied large language models to generate personalised recommendations and summaries, then **evaluated outputs against defined quality criteria** to refine prompt design and improve analytical quality.
- Iteratively tested and refined AI-driven recommendation workflows using user feedback and exploratory analysis, improving recommendation relevance.
- Built engagement and productivity metrics that quantified user behaviour and informed data-driven product decisions, communicated to stakeholders through dashboards and reports.

Python · SQL · Machine Learning · LLMs · pandas · PostgreSQL · AWS

Machine Learning Engineer — Freelance / Contract *Jan 2025 – Dec 2025*

- Delivered applied machine learning and analytics across multiple private-sector engagements, translating business questions into models and measurable recommendations.
- **Selected engagements —** customer analytics, churn modelling, predictive modelling, demand forecasting and behavioural analysis.
- Built end-to-end analytical workflows — data preparation, feature engineering, model training and evaluation — in Python and SQL.
- Deployed and maintained production models with Docker on AWS/GCP, automating testing and model versioning via GitHub Actions and MLflow.

Python · scikit-learn · SQL · AWS · GCP · Docker · GitHub Actions · MLflow

Data Analyst — Plan-Net, Leicester *Oct 2023 – Nov 2024*

- Automated recurring reporting with SQL and Python, reducing manual effort and improving turnaround for business stakeholders.
- Designed and maintained ETL workflows that improved data quality, consistency and reliability across reporting pipelines.
- Partnered with cross-functional stakeholders to translate operational data into clear reports and recommendations that supported decision-making.
- Analysed infrastructure and operational data to surface trends, recurring issues and opportunities for optimisation.

SQL · Python · ETL · dbt · Snowflake · Git

SELECTED PROJECTS (FULL CODE: GITHUB.COM/KNDUKUBA17-HUB)

Customer Churn Prediction · *Python, scikit-learn, XGBoost, SHAP*

- Predicted churn on **7,043 real telecom customers**; Logistic Regression achieved **ROC-AUC 0.842** (5-fold CV 0.845), 0.78 churn recall.
- Optimised for recall/ROC-AUC over accuracy; targeting the top 20% risk reaches **~50% of churners**.
- Handled class imbalance and explained drivers with coefficients and SHAP.

Customer Lifetime Value (CLV) Prediction · *Python, XGBoost, SHAP*

- Modelled 6-month customer spend from RFM behaviour on **1.07M real transactions** (UCI Online Retail II) with a leakage-safe temporal split.
- XGBoost reached test R^2 0.69 / MAE £600; the **top 10% predicted customers captured ~63% of actual future revenue** (vs 10% at random).
- Ranked the behavioural drivers of value with SHAP to make the output actionable for marketing.

Customer Churn Dashboard (interactive BI) · *JavaScript, SVG, HTML*

- Built an interactive retention dashboard on **7,043 real customers** with live client-side cross-filtering, KPI tiles (churn rate, monthly revenue at risk) and hand-built charts — no libraries.
- **Live:** kndukuba17-hub.github.io/Customer-Churn-Dashboard

E-Commerce Recommendation Engine · *Python, scikit-learn*

- Item-based collaborative filtering on **805k real transactions**, evaluated on a temporal hold-out with precision@k / recall@k.
- Beat a popularity baseline by **~1.5× on precision@10** (35% hit-rate) — a defensible offline evaluation.

EDUCATION

MSc, Applied Data Science & Cyber Security — Royal Holloway, University of London *2025 – 2026*
Modules: Machine Learning, Neural Networks, Big Data Analytics, Statistical Modelling, Cyber Threat Intelligence.

BSc, Psychology — University of Leicester *2021 – 2024*
Modules: Quantitative Research Methods, Behavioural Statistics, Cognitive Neuroscience.

A-Levels — Langley Park School *2019 – 2021*
Modules: Chemistry (A), Biology (A), Psychology (A).

Hobbies

Sports & Strategy: Fascinated by game theory and the analytical side of high-performance team sports.
Reading & Development: Avid reader of psychology and financial strategy (e.g., *Atomic Habits*, *Rich Dad Poor Dad*).
Music & Fashion: Deeply passionate about the intersection of design, sound, and modern culture; enjoy dissecting industry trends and applying mechanics to personal pursuits.